

Jeycar Calderón Badilla

Electromechanical Engineer

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Professional Profile

Electromechanical Engineer with experience in technical-financial analysis, mechanical and electrical systems, and engineering design. Focused on railway engineering, transportation infrastructure, and the transition toward clean energy solutions, with additional experience in HVAC control systems.

Academic Experience and Projects

Bachelor's Thesis: *Technical and Financial Prefeasibility Study on the Life Cycle Cost of Trains in Costa Rica for a Transition to an Electric Light Train.* [Link to document](#)

Tecnológico de Costa Rica (2024)

- Comparative evaluation between diesel and electric trains, with emphasis on railway maintenance and reliability.
- Preliminary study of electrical infrastructure and complete train life cycle.
- Analysis of environmental impact, sustainability, and energy savings in the railway sector.

International and Academic Participation

- Participation in the Global Mobility Call Congress (Madrid, 2024), focused on sustainable and railway mobility.
- Member of a research project focused on assessing railway infrastructure and preparing a technical report on its current condition and development needs.
- Contributing to the technical assessment of Costa Rica's electric railway project as part of a multidisciplinary research group at the Costa Rican professional association of engineers and architects (CFIA).
- Presented the fundamentals of railway electrification and electric traction systems to Electrical Engineering students in University of Costa Rica (UCR).

Work Experience

Johnson Controls

Application Engineer I in HVAC Control Systems

Apr 2025 - Present

- Development of controls solutions in HVAC systems, applying energy efficiency and reliability criteria.

Internship in HVAC Control Systems

Sep 2024 - Feb 2025

- Support in the implementation of control and monitoring projects for critical systems.

Education

Bachelor's Degree in Industrial Maintenance Engineering

Tecnológico de Costa Rica - 2024

Certification: Exploring The World Of Electric Buses: Advancing Zero-Emission Public Transport

TUMI, GIZ - 2025

Skills

Technical: Railway engineering, life cycle cost analysis (LCC), project management, ETAP, SolidWorks, AutoCAD, SCADA, Microsoft 365.

Soft: Leadership, multidisciplinary teamwork, strategic vision, effective communication, innovation, critical thinking.

Languages

Spanish: Native

English: Advanced (B2)

French: Basic (A1)